

TOLU OLATUJI

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Education

University of Toronto

Expected Graduation: April 2029

Bachelor of Applied Science in Computer Engineering — *PEY Co-op*

105 St. George Street, Toronto, ON

- Courses: Computer Fundamentals, Engineering Strategies and Practice, Introduction to Electrical and Computer Engineering, Electrical Fundamentals, Calculus II. Dynamics, Linear Algebra

Work Experience

University of Toronto

May 2025 – Present

Undergraduate Research Assistant — *Generative AI for Thermal Systems*

Toronto, ON

- Engineered a **Python** data pipeline using **OpenCV**, **Tesseract OCR**, and **SciPy** to extract values from **45+ CFD simulation images**, generating structured multi-modal **PyTorch tensors** for generative AI model training.
- Developing a supervised **PyTorch** deep learning model to predict temperature fields from velocity field inputs, creating a low-fidelity surrogate for **physics-informed neural networks (PINNs)** to accelerate design optimization.
- Applying data augmentation techniques to expand limited paired CFD simulation datasets, enabling training of **diffusion and GAN models** for automated battery cooling layout generation under geometric and manufacturing constraints.

Cornell University, BTI Institute

May 2024 – January 2025

Research Intern

Ithaca, NY, US

- Evaluated phenotypic expression across **35,000 CNR (Colorless Non-Ripening) tomato line samples** in a gene mutation analysis study, generating critical data that identified genetic triggers for fruit maturation.
- Processed large-scale biological datasets using Python and Excel to extract, filter, and summarize relevant features, improving the accuracy and efficiency of downstream data modeling and analysis pipelines.
- Synthesized experimental data into comprehensive technical reports supporting collaborative research and publications.

Projects

Questr

Oct 2025 – Nov 2025

Hackathon Project – *NewHacks (University of Toronto)*

- Developed a full-stack web app using **Flask (Python)**, **SQLite**, and **Bootstrap 5** to deliver personalized daily wellbeing quests powered by the **Google Gemini API**.
- Built a sentiment analysis module with **TextBlob** to evaluate user reflections and visualize trends using **Chart.js**.
- Deployed on **Render.com** using Gunicorn with environment-based configuration for secure, production-ready hosting.

Custom PCB Project

December 2025

Altium Designer, PCB Layout, DFM

- **Schematic Design & System Definition:** Created a complete PCB project from concept, developing organized schematics with properly defined components, net classes, and electrical constraints using custom and vendor libraries.
- **PCB Layout & Design Rules:** Performed component placement and multi-layer PCB routing, configuring stack-up parameters, copper pours, and design rules to ensure electrical integrity and manufacturability.
- **Manufacturing Readiness & Outputs:** Prepared the design for fabrication by generating industry-standard **Gerber**, **NC drill**, and **pick-and-place** files, along with a complete **BOM** suitable for automated PCB assembly.

Extracurriculars

University of Toronto Aerospace Team (UTAT) | *Avionics Team Member*

August 2025 – Present

- Engineering a **Ullage Capacitive Fill Sensor** to monitor propellant levels in microgravity, using **Altium Designer**.
- Developing embedded firmware on **STM32 microcontrollers** to process low-capacitance signals, aiming to ensure high-precision data telemetry for critical flight-control systems.

Technical Skills

Languages / Tools: Python, Java, JavaScript, C, SQL, HTML5, CSS, Git, VSCode, Jupyter, STM32, Altium

Libraries / Frameworks: PyTorch, NumPy, OpenCV, SciPy, Matplotlib, Flask, Pytesseract, Bootstrap 5, Render